Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 26)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

0 1 . 1 Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

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0 1 . 2 Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

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0 1 . 3 Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

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0 2 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 2 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 2 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 3 . 1 Complete Table 1 to show a suitable sampling technique for each group of organisms and one limitation of the technique.

[6 marks]

Table 1

Organisms	Technique	One limitation	
Invertebrates active on the ground surface at night			
Invertebrates living in leaf litter			
Invertebrates on tree branches			

Do not write outside the box

0 3 . 2 Explain why a light trap cannot be used to prove that a moth species is absent from a site.

[2 marks]

0 3 . 3 Explain how a motor-driven suction sampler can give a population density, whereas a pooter cannot.

[2 marks]

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0 4 . 1 Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

0 4 . 2 Explain how pheromone traps can be used in pest control.

[2 marks]

0 4 . 3 Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

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0 4 . 4 Give **one** way farmers can support crop pollinators.

[1 mark]

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0 5 Table 1 shows features of some fishing methods.

0 5 . 1 Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

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0 5 . 2 Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

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0 5 . 3 Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

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0 6 . 1 Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

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Satellite / radio / GPS tracking .....

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DNA databases and eDNA .....

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Image recognition software .....

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Acoustic monitoring / sonograms .....

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0 6 . 2 Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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0 7 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 7 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 7 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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0 8 . 1 Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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0 8 . 2 Describe **five** features of a suitable release site.

[5 marks]

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0 8 . 3 Explain the importance of post-release monitoring.

[2 marks]

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0 9 . 1 Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

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0 9 . 2 Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

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0 9 . 3 Describe **three** threats to UK broadleaf woodland.

[3 marks]

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1 0 . 1 Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

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1 0 . 2 Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

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1 0 . 3 Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

1 0 . 4 Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

11 . 1 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

OR

11 . 2 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS